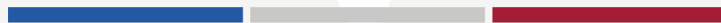




MIT SLOAN SCHOOL OF MANAGEMENT
MIT SCHWARZMAN COLLEGE OF COMPUTING



MODULE 1 UNIT 1
Lesson Video 4 Transcript

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Introduction

PAUL MCDONAGH-SMITH: We're very pleased to have Kanch Patlolla here with us today. Kanch, it's a pleasure to have you joining our program. Would you like to take a moment to introduce yourself to our group?

KANCHANA PATLOLLA: Hi, Paul. Thanks for having me here. It is always a great pleasure to be with this kind of group. I'm Kanch. I'm a product solutions manager at Google, and I have been with Google for nine years, close to nine years now. Pretty much in the AI journey adoption, but for the past three years in this generative AI world of things. I particularly focus in the agents side of the world. And, in my day-to-day job, I kind of meet close to, like, 8–10 customers a week on their journey, adoption to generative AI.

What is an agent?

MCDONAGH-SMITH: Kind of focusing in on those applications, in your experience and work that you've been doing over the last period, are there particular types of business problems or workflows that you're finding are especially well suited to Agentic AI? And where do you see the greatest ROI potential right now in this period?

PATLOLLA: When it comes to agents, agents conversations started in November of 2023 with React, which is reasoning and acting paper. That's where it started.

MCDONAGH-SMITH: Yep.

PATLOLLA: And then when we went to chain of thought prompting paper, and then now we are talking about, you know, other recursive models, etc.

So, in this journey, the definition of agent has increased by itself. Like, what is an agent used to be the model with the ability to call a tool. That is all it was. And it needs to be able to reason and act. Like, the reasoning is between these ten tools, which tool would I be able to call? That is where it stopped, where it was. And then we introduced this powerful memory to it. Where now we had the ability to provide more stateful context that helped the agent to do the reasoning well and to perform well. And then they introduced the concept of multi-agents. So the unit of work was, you know, separate. So we were creating specialized agents and it was going. That was the second phase of it, I would say. But now more things are coming that, I think I would say GPT3, they released with the reasoning model. So the model has reasoning. So it's the evolution of where the models and agents come, particularly how you develop, it was changing. Because what you thought as agents is now changed to now models, because that became the model ability.

So when I think about agent today, if you talk to me about today, the reasoning ability is now in the model. The memory ability, particularly with the skills from Claude and OpenAI, is now in the model back itself. And model does have the ability to call tools because it was always function calling and tool calling was there. So all of these now exist in the model.

So now what is an agent? So now we need to redefine what an agent is. So when you think about agents, the interesting thing is the plumbing work in my opinion.

MCDONAGH-SMITH: Yeah. Okay.

PATLOLLA: Because it's the infrastructure, the ability to orchestrate, the ability to, I would even call it as a deployable model is an agent more. The reason why I went through that journey. Sorry.

MCDONAGH-SMITH: No, it's really interesting and very valuable actually to see that transition, that evolution perhaps.

PATLOLLA: The adoption. Yeah. The reason why I went through that is the definition has changed over time. So I don't know if there is a good ROI agent use case. There are several agents, which are implemented, and there are several model-based agents which are implemented. And most of them are good at certain things. Like, they are good at, you know I have seen several design patterns evolve. I have seen RAG-based, which is retrieval-based-augmentation-based agents come into picture. I have seen a master, which is an orchestrator, and then it has several, you know, small agents that defines and then specialized ways. And I've seen, you know, parallel agents running so that the latency is reduced in some instances. I see several of those patterns definitely. But when I think about an AI adoption, particularly with respect to ROI, I think there is a lot more which is coming, which will be more exciting than what we have today.

SPEAKER: Did you understand all the concepts covered in this video? If you'd like to review any of the content, please click on the chapter menu.